Practice runs and evidence

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use repeated trials to prepare an evidence-based design explanation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Ideal speed becomes 5 times as large. Actual rpm, load or slipping may change.

02. Worked example: explain the model
All three retained; mean≈2.17 s, fastest 2.0 s.

03. Read and interpret the model
Actual evidence with functional labels and explanations.

04. Apply the physics
0.80 m/s for supplied example; actual values must be measured.

05. Explain your reasoning
Retain and annotate failure; report completed-run speeds separately with inclusion criteria and failure count.

06. Review: connect an earlier idea *Review: Unit 7, Day 14.*
Faster flux change or more turns increases emf magnitude. Stronger field/effective area changes are valid with suitable controls.

07. Exit check
Student cites actual evidence and acknowledges if evidence is limited.

08. STRETCH: test the model
Second set faster mean 2.1 s and more consistent; first has fastest individual run 2.0 s. Choice depends on stated criteria.

Practice runs and evidence

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use repeated trials to prepare an evidence-based design explanation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Torque ratio 7; speed ratio 1/7. Assumes ideal gears with negligible losses.

02. Worked example: explain the model
All three retained; mean≈2.17 s, fastest 2.0 s.

03. Read and interpret the model
Actual evidence with functional labels and explanations.

04. Apply the physics
Mean 5.0 s; range 0.4 s; 0.80 m/s from mean time.

05. Explain your reasoning
Retain and annotate failure; report completed-run speeds separately with inclusion criteria and failure count.

06. Review: connect an earlier idea *Review: Unit 7, Day 14.*
Emf is 4 times as large. Same turn count and flux change per turn; compare average emf over each interval.

07. Exit check
Claim is bounded; numerical/observational evidence named; physical mechanism fits; causal certainty matches the comparison.

08. STRETCH: test the model
Mean speeds (1.5+1.0)/2=1.25 m/s; total 6/5=1.2 m/s. Equal-trial and time-weighted averaging differ.